

Md Mahfuzur Rahman

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Hardware and photonic IC engineer with strong communication skills; experienced in FDTD simulation, device fabrication, testing, and advanced packaging; enthusiastic learner with a proactive, problem-solving mindset.

EDUCATION

University of Florida, GPA: 3.71 (ongoing) Ph.D. in Electrical and Computer Engineering	Gainesville, FL Aug 2024 — Dec 2027
University of Florida, GPA: 3.61 M.S. in Electrical and Computer Engineering	Gainesville, FL Graduated December 16, 2025
Coursework VLSI 1 Electromagnetic Fields & Applications Fundamentals of Photonics Nano & Quantum Photonics App Microelectronic Fabrication Technologies Semiconductor Device Fabrication Lab Physical attacks and inspection of Electronics ML & Python	

SKILLS SUMMARY

EDA / Simulation	Ansyes Lumerical (FDTD, MODE, DEVICE), Synopsys RSoft, KLayout, Cadence (Virtuoso XL, DRC, LVS)
Fabrication	Photolithography, PECVD Deposition, Diffusion, Sputtering, ALD, Dry (ICP, DRIE) & Wet Etching
Testing & Metrology	SEM, FIB, Filmetrics, Profilometer, Probe Station Microscopy
Programming	Python, MATLAB, Verilog, Mathematica, ETX

PROFESSIONAL EXPERIENCES

UF Herbert Wertheim College of Engineering, University of Florida, Gainesville, FL Graduate Research Assistant	Aug 2024 — Present
• Developed and tested photonic ASIC components using ML-guided design and cleanroom fabrication, supported chip testing/packaging, and contributed to publications/presentations/patent filings.	
Lyle School of Engineering, Southern Methodist University, Dallas, TX Graduate Teaching Assistant	Aug 2023 — July 2024
• Facilitated lab sessions for the course, supporting students and improving lab completion and troubleshooting efficiency.	

RESEARCH EXPERIENCES

Photonic IC Research: Integrated Nano-Mirror Couplers (NMC) and Photonic Wire Bonding (PWB)	Aug 2024 — Present
This research focuses on developing scalable, low-loss vertical optical interconnects for chip-to-chip and wafer-scale packaging.	
• Optimized NMC geometry across multiple design variants, reducing loss by 0.17 dB and achieving ± 50 nm alignment tolerance.	
• Developing DOE-based process integration and failure analysis checklists, standardizing wafer-run execution and metrology.	
SRAM Design Project using Cadence Virtuoso	Aug 2024 — Dec 2024
This project aims to design an SRAM (Static Random Access Memory) architecture with optimized performance and reliability.	
• Simulated standard cells (INV/NAND/NOR) and SRAM blocks, validating functionality across PVT variations to ensure reliability.	
• Implemented 45nm FinFET designs in Virtuoso, completing I/O simulations and DRC/LVS to deliver signoff-ready layouts.	
Fabrication and Characterization of MOSFETs, HBTs, and Broad Area Lasers	Aug 2023 — May 2024
This project focuses on the fabrication of semiconductor devices.	
• Fabricated MOSFETs, HBTs, and broad-area lasers through multi-step cleanroom processing and device-level test preparation.	
• Measured I-V curves and summarized performance variability across samples for comparative evaluation.	

PATENTS & PUBLICATIONS

Patents

1. **Vertical and lateral optical three dimensional interconnect fabric**, Inventors: Volker J. Sorger, Hangbo Yang, Md Mahfuzur Rahman, United States Patent Trademark Office, 19/366,635
2. **Hollow Core Vertical Waveguides for Photonic Interconnects and Routing Fabrics**, Inventors: Volker J. Sorger, Hangbo Yang, Md Mahfuzur Rahman, United States Patent Trademark Office, 63/780,669

Journal

1. **In-substrate Nano-mirror Coupler and Hollow-core Vertical Waveguides for High-density 3D Photonic Routing Fabrics**
Md Mahfuzur Rahman, Hangbo Yang, Hao Wang, Volker J. Sorger JLT'26

Conference

1. **Optimizing Nano-Mirror Coupler Design for High-Efficiency Optical Interconnects**
Md Mahfuzur Rahman, Hangbo Yang, Nicola Peserico, Hao Wang, Volker J. Sorger SPIE'25

AWARDS & CERTIFICATIONS

Standing InnOvation Award Photonic Integrated Circuit – Edx	UF Innovate Tech Licensing (2025) MIT
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